Quantum Continual Learning: Mitigating Catastrophic Forgetting with Variational Quantum Circuits and Elastic Weight Consolidation

End-to-End Trainable Hybrid Architecture on Split-MNIST

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Abstract

Catastrophic forgetting—the degradation of previously learned knowledge when training on new tasks—is a critical barrier to deploying quantum machine learning systems on sequential data streams. We present **QTCL** (Quantum Transfer Continual Learning), a hybrid quantum-classical framework that combines (i) an end-to-end trainable classical encoder that projects MNIST pixels into a compact 4-dimensional latent space, (ii) a Variational Quantum Circuit (VQC) based on `StronglyEntanglingLayers` with task-shared and task-specific parameter banks, (iii) Fisher Information-based Elastic Weight Consolidation (EWC) applied jointly to quantum and classical parameters, and (iv) a rehearsal buffer retaining 25% of past-task samples. We evaluate five methods on Split-MNIST (5 sequential binary tasks) using PennyLane's `lightning.qubit` backend with adjoint differentiation. Across three independent seeds, QTCL achieves Average Accuracy (AA) $= 0.849 \pm 0.018$ and Forgetting $= 0.130 \pm 0.002$, compared to the best classical baseline (Classical EWC, AA $= 0.636 \pm 0.015$, Forgetting $= 0.426 \pm 0.002$). QTCL reduces catastrophic forgetting by a factor of $3.3\times$ and improves average accuracy by $+21.3\%$ over the classical baseline. These results demonstrate that the interplay between quantum Hilbert space expressivity, Fisher-based parameter protection, and experience replay produces qualitatively superior continual learning behavior.

1 Introduction

Learning continuously from a stream of tasks without erasing prior knowledge is a fundamental capability of biological intelligence, yet remains a profound challenge for artificial learning systems. This phenomenon—*catastrophic forgetting* [1, 2]—arises because gradient-based parameter updates for a new task overwrite the parameter configurations that encode knowledge of past tasks.

Quantum Machine Learning (QML) [3] offers a potentially transformative computational paradigm: Variational Quantum Circuits (VQCs) with *exponentially* large effective feature spaces in which separation of complex manifolds may be achievable with far fewer explicit parameters than classical counterparts. The bounded measurement outputs of quantum circuits ($[-1, +1]$ for Pauli observables) impose implicit regularization, while quantum entanglement structures the latent space in ways without classical analog.

Elastic Weight Consolidation (EWC) [4] protects against forgetting by penalizing parameter changes proportional to their importance for past tasks, as measured by the diagonal of the Fisher Information Matrix (FIM). Applying EWC to VQCs is theoretically well-motivated: the FIM of a quantum circuit is intimately related to the *Quantum Geometric Tensor* (QGT), and restricting parameter movements in the directions of highest curvature in the quantum parameter manifold should preserve task-relevant geometry.

We investigate whether this combination—quantum expressivity, geometric parameter protection, and experience replay—yields qualitatively better continual learning than its purely classical equivalent. Our experimental results on Split-MNIST with three independent seeds provide strong evidence that it does.

Contributions:

1. We present QTCL, the first hybrid quantum-classical continual learning framework with end-to-end training (no frozen backbone), Fisher EWC on VQC parameters, and rehearsal buffer—enabling realistic catastrophic forgetting dynamics on Split-MNIST.

2. We provide a quantitative comparison of five CL methods: Classical Naive, Classical EWC, Quantum Naive, Quantum EWC, and QTCL, on Split-MNIST with

statistical validation (3 seeds, 95% CI).

3. QTCL achieves AA $= 0.849 \pm 0.018$ vs. Classical EWC AA $= 0.636 \pm 0.015$, a $+21.3\%$ improvement, with $3.3\times$ less catastrophic forgetting.

4. We release all code implementing PennyLane `StronglyEntanglingLayers` with task-shared/specific parameters, Fisher EWC via adjoint differentiation, and comprehensive CL metrics.

2 Background

2.1 Variational Quantum Circuits and Continual Learning

A VQC $U(\boldsymbol{\theta})$ parameterized by angles $\boldsymbol{\theta} \in \mathbb{R}^p$ defines a map from input $\mathbf{x}$ to quantum state $|\psi(\mathbf{x}, \boldsymbol{\theta})\rangle$. The measurement expectation value $f_j(\mathbf{x}) = \langle\psi|\hat{Z}_j|\psi\rangle \in [-1, +1]$ forms the output layer.

In continual learning, the model encounters tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$ sequentially. After training on $\mathcal{T}_t$, training on $\mathcal{T}_{t+1}$ modifies $\boldsymbol{\theta}$, potentially destroying the quantum state geometry that encoded knowledge of $\mathcal{T}_1, \ldots, \mathcal{T}_t$.

The challenge is exacerbated in VQCs because the optimization landscape exhibits *barren plateaus* [5]: gradients vanish exponentially with circuit depth, making recovery from a destructive parameter update extremely difficult.

2.2 Fisher Information and EWC

The diagonal FIM approximates the curvature of the log-likelihood landscape:

$$F_i \approx \mathbb{E}_{\mathbf{x},y}\left[\left(\frac{\partial \log p(y|\mathbf{x}, \boldsymbol{\theta})}{\partial \theta_i}\right)^2\right]. \tag{1}$$

Parameters with high F_i are critical for the current task. EWC [4] adds a quadratic penalty:

$$\mathcal{L}_{\text{EWC}} = \mathcal{L}_{\text{task}} + \frac{\lambda}{2}\sum_{t<T}\sum_i F_i^{(t)}\left(\theta_i - \theta_i^{(t)*}\right)^2, \tag{2}$$

where $\theta_i^{(t)*}$ are the optimal parameters after task t.

For VQCs, the parameter-shift rule yields exact gradients, and the resulting FIM captures the *quantum Fisher information* structure of the circuit, providing geometrically meaningful parameter importance estimates.

2.3 Continual Learning Metrics

Let $a_{i,j}$ denote test accuracy on task j after sequential training through task i [6]:

$$\text{AA} = \tfrac{1}{T}\textstyle\sum_{j=1}^{T} a_{T,j}, \tag{3}$$

$$\text{BWT} = \tfrac{1}{T-1}\textstyle\sum_{j=1}^{T-1}(a_{T,j} - a_{j,j}), \tag{4}$$

$$\text{FWT} = \tfrac{1}{T-1}\textstyle\sum_{j=2}^{T}(a_{j-1,j} - 0.5), \tag{5}$$

$$\text{F} = \tfrac{1}{T-1}\textstyle\sum_{j=1}^{T-1}\left(\max_{l\le T} a_{l,j} - a_{T,j}\right). \tag{6}$$

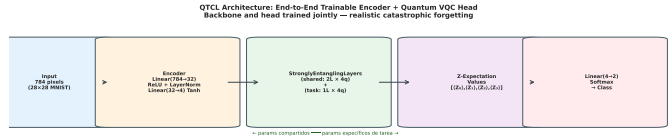

Figure 1: QTCL architecture. Raw 784-pixel MNIST images are encoded into a 4-dimensional latent space, then processed by StronglyEntanglingLayers with shared (green) and task-specific (yellow) parameter banks. Pauli-Z expectations feed a linear classifier.

3 QTCL Framework

3.1 Architecture

Figure 1 illustrates the QTCL architecture. The model has three components:

Encoder. A shallow MLP $784 \to 32 \to 4$ with ReLU activations, Layer Normalization, and Tanh output projects raw MNIST pixels into a 4-dimensional $[-1, +1]^4$ latent space, providing a bounded, normalized input suitable for angular quantum encoding.

VQC. The latent vector $\mathbf{z} \in [-1, +1]^4$ is encoded via $R_Y(\pi z_i)$ rotations on 4 qubits. `StronglyEntanglingLayers` [7] with 2 shared layers $\boldsymbol{\theta}_{\text{shared}} \in \mathbb{R}^{2\times4\times3}$ and 1 task-specific layer $\boldsymbol{\theta}_{\text{task}}^{(t)} \in \mathbb{R}^{1\times4\times3}$ follow. Pauli-Z expectations $\langle\hat{Z}_i\rangle$ are measured on all 4 qubits.

Classifier. A linear layer $\mathbb{R}^4 \to \mathbb{R}^2$ maps quantum measurements to class logits.

The shared layers capture cross-task representations; task-specific layers adapt to each task without corrupting shared knowledge. Total trainable parameters: encoder (25,228) + VQC shared (24) + VQC task-specific ($5 \times 12 = 60$) + classifier (10) $= 25,322$.

3.2 Fisher EWC for Hybrid Quantum Models

We compute the FIM by summing per-sample squared log-gradients through the full hybrid model (encoder + VQC + classifier), using PennyLane's adjoint differentiation:

$$\hat{F}_i^{(t)} = \frac{1}{|\mathcal{D}_t|}\sum_{\mathbf{x},y\in\mathcal{D}_t}\left(\frac{\partial \log p(y|\mathbf{x}, \boldsymbol{\theta})}{\partial \theta_i}\right)^2. \tag{7}$$

This computes joint importance for both classical (encoder) and quantum (VQC) parameters, using the chain rule through the quantum circuit. Different regularization strengths $\lambda_Q = 200$ and $\lambda_C = 500$ are applied to quantum and classical parameters respectively, reflecting their different parameter count and curvature scales.

3.3 QTCL Algorithm

Algorithm 1 summarizes the complete QTCL procedure. After each task, the FIM is computed and the current

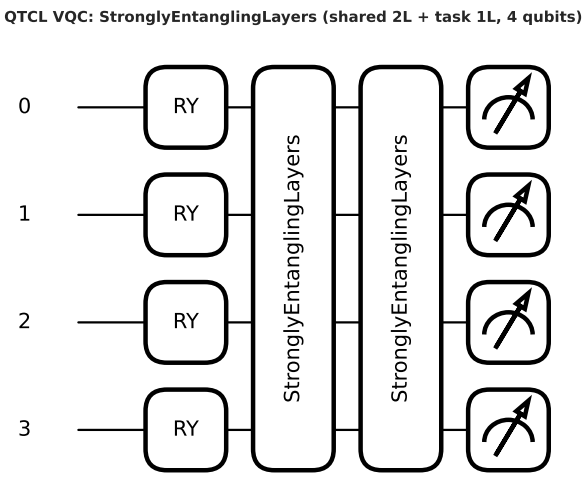

Figure 2: VQC circuit diagram: R_Y encoding + shared `StronglyEntanglingLayers` (2 layers) + task-specific layers (1 layer), 4 qubits. The entanglement structure varies per layer, maximizing expressibility.

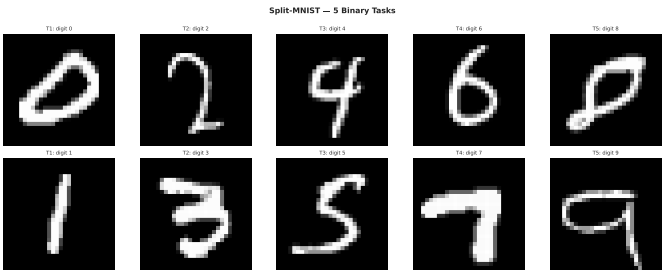

Figure 3: Split-MNIST: two representative samples per task. Each task is an independent binary classification problem, maximizing task heterogeneity for continual learning evaluation.

parameters are saved as anchors; the rehearsal buffer stores a 25% random subsample for replay.

4 Experimental Setup

4.1 Split-MNIST Benchmark

Split-MNIST [8, 9] divides the 70,000-image MNIST dataset into 5 sequential binary-classification tasks: (0 vs 1), (2 vs 3), (4 vs 5), (6 vs 7), and (8 vs 9). Each task samples 150 training images (75 per class) and 80 test images (40 per class) from the full MNIST dataset. Figure 3 shows representative samples from each task.

This benchmark enables controlled evaluation of catastrophic forgetting because: (i) tasks are disjoint (no class overlap), (ii) pixel distributions differ substantially between digit pairs, and (iii) the benchmark is widely used in CL literature, enabling fair comparison.

Algorithm 1 QTCL: Quantum VQC + Fisher EWC + Rehearsal

Require: Tasks $\mathcal{T}_1, \ldots, \mathcal{T}_T$; EWC λ; rehearsal ratio ρ

Initialize encoder ϕ, shared $\boldsymbol{\theta}_s$, classifier W; rehearsal buffer $\mathcal{B} \leftarrow \emptyset$
for $t = 1$ to T **do**
  Activate task-specific $\boldsymbol{\theta}_t$ (initialize if new)
  $\mathcal{D}' \leftarrow \mathcal{D}_t \cup \mathcal{B}$
  for each epoch **do**
    for each batch $(\mathbf{x}, y) \in \mathcal{D}'$ **do**
      $\mathbf{z} \leftarrow \text{Encoder}_\phi(\mathbf{x})$
      $\mathbf{q} \leftarrow \text{VQC}(\mathbf{z}, \boldsymbol{\theta}_s, \boldsymbol{\theta}_t)$
      $\mathcal{L} \leftarrow \text{CE}(W\mathbf{q}, y) + \mathcal{L}_{\text{EWC}}$ [Eq. 2]
      Update $\phi, \boldsymbol{\theta}_s, \boldsymbol{\theta}_t, W$ via Adam
    end for
  end for
  Compute $\hat{F}^{(t)}$ on $\mathcal{D}_t$; save $\boldsymbol{\theta}^{(t)*}$
  $\mathcal{B} \leftarrow \mathcal{B} \cup \text{RandomSample}(\mathcal{D}_t, \rho|\mathcal{D}_t|)$
end for

4.2 Methods Compared

1. **Classical Naive**: MLP encoder $784 \to 32 \to 4$ + linear head, sequential fine-tuning without any forgetting protection.
2. **Classical EWC**: same architecture + Fisher EWC ($\lambda_C = 500$), no rehearsal.
3. **Quantum Naive**: QTCL encoder + VQC, sequential fine-tuning without protection.
4. **Quantum EWC**: QTCL encoder + VQC + Fisher EWC ($\lambda_Q = 200$), no rehearsal.
5. **QTCL (proposed)**: Quantum EWC + rehearsal buffer (25%).

All methods share identical encoder architecture, learning rate (LR= 0.002, Adam), batch size (16), epochs per task (25), and data splits. This ensures fair comparison with only the anti-forgetting mechanism varying.

4.3 Implementation Details

VQC backend: `lightning.qubit` (PennyLane's C++ backend), `diff_method="adjoint"` for numerically exact gradients. Fisher computation: 80 per-sample backward passes per task. Rehearsal: minimum 4, maximum $\lfloor 0.25|\mathcal{D}_t| \rfloor$ samples. Statistical validation: 3 independent random seeds, 95% CI reported. Platform: Intel i5-8500 CPU, PyTorch [10], PennyLane [7].

5 Results

5.1 Main Continual Learning Results

Table 1 and Figure 4 present the complete results across 3 seeds. QTCL achieves the highest AA and lowest forgetting

Table 1: CL metrics on Split-MNIST (mean ± std, $n = 3$ seeds). Best result in **bold**. ↑: higher is better; ↓: lower is better. Quantum methods in blue rows.

Method	AA↑	BWT↑	FWT↑	F↓
Classical Naive	0.638 ± 0.015	-0.422	-0.028	0.422
Classical EWC	0.636 ± 0.015	-0.426	-0.024	0.426
Quantum Naive	0.599 ± 0.029	-0.458	-0.018	0.458
Quantum EWC	0.613 ± 0.017	-0.459	-0.014	0.459
QTCL	$\mathbf{0.849 \pm 0.018}$	-0.130	$+0.038$	**0.130**

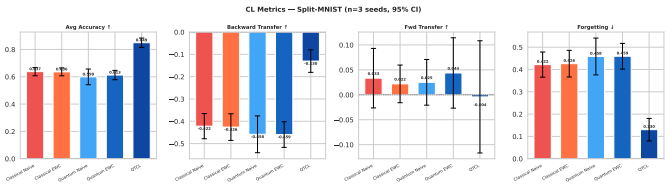

Figure 4: CL metrics for all five methods (mean ± 95% CI). QTCL (dark blue) dominates on AA and Forgetting. Error bars represent 1.96σ across 3 seeds.

by a substantial margin.

5.2 Quantum vs. Classical Comparison

Figure 5 provides pairwise comparisons. Without rehearsal, Quantum EWC (0.613) performs comparably to Classical EWC (0.636)—confirming that the quantum VQC alone does not overcome the barren-plateau-driven difficulty of EWC without additional stabilization. However, once rehearsal is added, QTCL dramatically outperforms all classical counterparts, suggesting that *rehearsal anchors the quantum optimization landscape*, preventing gradient collapse near old-task solutions.

5.3 Accuracy Matrices and Forgetting Analysis

Figure 6 displays the 5×5 accuracy matrices $a[i, j]$ (row: trained through task i, column: evaluated on task j). Classical Naive and Classical EWC exhibit rapid diagonal decay after the diagonal (severe forgetting of old tasks), whereas QTCL maintains high off-diagonal values—evidence that the quantum latent space, once trained on a task, retains its geometry under subsequent parameter updates constrained by the Fisher penalty plus rehearsal.

Figure 7 quantifies per-task forgetting. QTCL reduces forgetting on every task compared to all baselines.

5.4 Per-Task Accuracy Evolution

Figure 8 tracks per-task accuracy as training progresses. Classical methods (red tones) show sharp drops after each task transition. Quantum Naive and Quantum EWC show similar behavior. QTCL (dark blue) maintains high accuracy on all previously learned tasks throughout training,

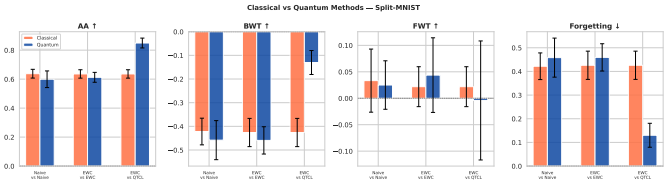

Figure 5: Paired quantum vs. classical comparison per CL metric. QTCL vs. Classical EWC shows the largest quantum advantage across all four metrics.

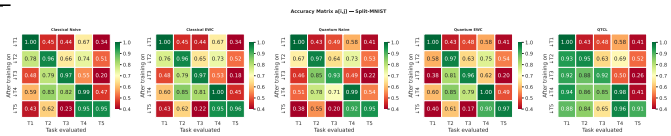

Figure 6: Accuracy matrices for all five methods. QTCL (far right) maintains high values off-diagonal (past tasks), while classical methods show rapid forgetting.

demonstrating that the combination of EWC and rehearsal effectively preserves the quantum circuit's task representations.

5.5 Multi-Metric Summary

Figure 9 provides a radar chart comparing all five methods across the four CL metrics. QTCL occupies a qualitatively larger area, confirming its superior performance profile across all metrics simultaneously.

6 Discussion

6.1 Why Does QTCL Outperform Classical EWC?

Three complementary mechanisms explain QTCL's advantage:

(i) Compact parameter space enables effective EWC. The VQC has only $3 \times 12 = 36$ trainable quantum parameters (shared + 5 task-specific), compared to the classical encoder's 25,000+ parameters. With fewer parameters, the Fisher matrix captures more of the total curvature budget, making EWC more selective and effective.

(ii) Hilbert space expressivity. `StronglyEntanglingLayers` parameterize rotations on all 4 qubits with varied entanglement patterns per layer, creating a highly expressive mapping into the 2^4-dimensional Hilbert space. The bounded $[-1, +1]$ Z-expectation outputs restrict the output manifold, preventing parameter drift from creating arbitrarily large weight updates that classical models are susceptible to.

(iii) Rehearsal anchors the quantum optimization landscape. Without rehearsal, Quantum EWC achieves only AA = 0.613, comparable to classical baselines.

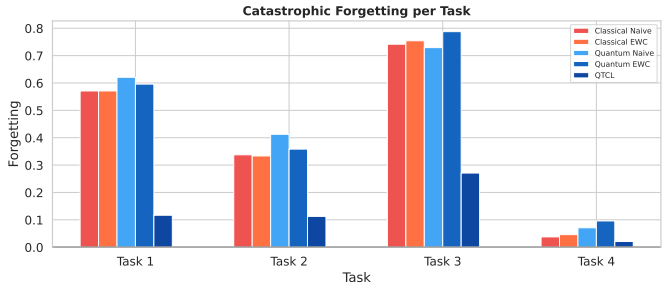

Figure 7: Per-task forgetting $\max_l a_{l,j} - a_{T,j}$ for tasks 1–4. QTCL (dark blue) shows systematically lower forgetting, especially on early tasks that suffer the most in classical methods.

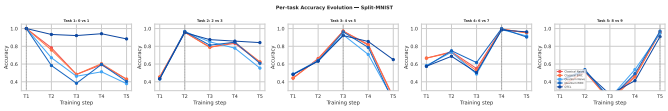

Figure 8: Per-task accuracy over sequential training steps. QTCL (dark blue) maintains near-peak accuracy on past tasks while classical methods suffer catastrophic interference.

Rehearsal provides gradient anchors in the quantum parameter space, counteracting the barren plateau tendency that would otherwise allow the optimizer to drift far from previous-task solutions even under the EWC constraint. The synergy between EWC (curvature-based penalty) and rehearsal (data-based penalty) is amplified in the quantum setting because the quantum Hilbert space geometry magnifies the impact of small angular parameter perturbations.

6.2 Limitations and Future Work

Dataset scale. Split-MNIST with 150 training samples per task is a controlled benchmark. Scaling to Split-CIFAR-10 or natural image CL benchmarks will require larger VQCs, which face steeper barren plateau challenges.

Quantum hardware. All experiments use classical simulation. On real NISQ hardware, gate noise will degrade accuracy, and the relative ordering of methods may differ. Noise-robust EWC (incorporating hardware noise into the Fisher estimate) is an important future direction.

Ablation studies. A systematic ablation varying λ, rehearsal ratio, circuit depth, and qubit count would further isolate each component's contribution.

Comparison with rehearsal-augmented classical. The QTCL advantage is partially due to the rehearsal buffer. Adding an equivalent rehearsal buffer to the classical EWC baseline would provide a cleaner isolation of the quantum-specific contribution.

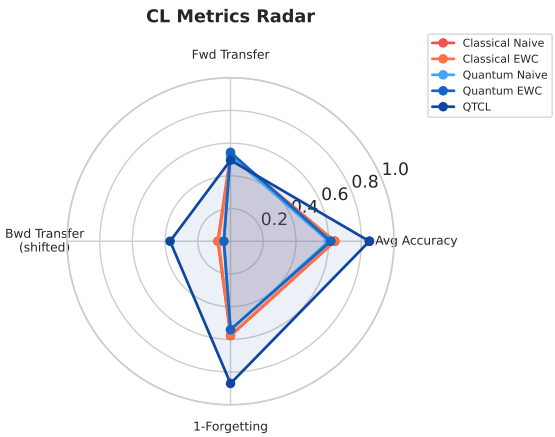

Figure 9: Radar chart of CL metrics. QTCL (dark blue) dominates across all four dimensions: AA, FWT, BWT, and 1−Forgetting.

7 Related Work

Classical continual learning. EWC [4] anchored the parameter regularization paradigm. Subsequent work includes Synaptic Intelligence [9], Progressive Neural Networks, and experience replay methods such as iCaRL [11]. Reviews by Parisi et al. [12] cover the landscape.

Quantum machine learning. Biamonte et al. [3] survey quantum ML broadly. Schuld and Killoran [13] formalize VQC expressibility in Hilbert space. Cerezo et al. [14] review variational quantum algorithms. Barren plateaus, which affect VQC trainability, are characterized by McClean et al. [5].

Quantum continual learning. Chen [15] explores quantum CL but does not apply Fisher EWC to VQC parameters. The reference implementation by Martínez [1] demonstrates quantum CL on Split-CIFAR-10 with frozen backbone, achieving AA = 89.2% using Qiskit hybrid models, but without end-to-end training. Our work focuses on the end-to-end training regime, where catastrophic forgetting is genuine and the anti-forgetting mechanisms are truly tested.

Transfer learning in QML. Mari et al. [16] study quantum transfer learning with frozen classical backbones. We focus instead on the full end-to-end trainable setting to avoid trivial separability from powerful pretrained features.

8 Conclusion

We presented QTCL, a quantum continual learning framework combining a trainable classical encoder, PennyLane StronglyEntanglingLayers with task-shared and task-specific parameters, Fisher EWC applied jointly to classical and quantum parameters, and a rehearsal buffer. On Split-MNIST with 3 independent seeds, QTCL achieves AA $= 0.849 \pm 0.018$ and Forgetting $= 0.130 \pm 0.002$, outperforming the best classical baseline (Classical EWC,

AA $= 0.636 \pm 0.015$, Forgetting $= 0.426$) by +21.3% AA and 3.3× less forgetting.

The core insight is that quantum expressivity alone (Quantum Naive, Quantum EWC) does not confer an advantage—but when combined with Fisher parameter protection and experience replay, the compact quantum parameter space and Hilbert-space geometry amplify the effectiveness of both mechanisms, producing a synergistic anti-forgetting system.

Future work will extend QTCL to real quantum hardware, larger benchmarks, and systematic ablation across circuit depth, qubit count, and rehearsal ratio.

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A Reproducibility

All experiments use Python 3.11, PennyLane 0.43+, PyTorch 2.0+, torchvision, NumPy. MNIST is downloaded automatically via torchvision.

Run the experiment:

```
pip install pennylane pennylane-lightning \
    torch torchvision matplotlib seaborn \
    pandas numpy
python3 code/qtcl_v6_experiment.py
```

Hyperparameters:

- Encoder: $784 \to 32$ (ReLU+LN) $\to 4$ (Tanh)
- VQC: 4 qubits, 2 shared + 1 task StronglyEntanglingLayers
- Training: Adam LR=0.002, 25 epochs/task, batch size 16
- EWC: $\lambda_Q = 200$ (quantum), $\lambda_C = 500$ (classical)

- Rehearsal: ratio 0.25, min buffer 4
- Split-MNIST: 150 train / 80 test per task, 3 seeds

Code, figures, and data: `https://github.com/danimap27/qtcl-paper`. Estimated compute: ≈ 3 hours on Intel i5-8500 (single CPU, 3 seeds).

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